Earnings Straddle Mispricing: A Cross-Sectional Alpha Study

Introduction

Earnings announcements are important information events for stocks. When a company reports earnings, investors learn new information about revenue, profits, margins, guidance, and future expectations. Because of this, stocks often move sharply after earnings. These moves can be positive or negative, but for options traders, the direction is not always the main concern. A trader who buys a straddle is mainly betting that the stock will move a lot, regardless of direction.

A straddle is an options position consisting of one call option and one put option with the same underlying stock, strike price, and expiration date. In this project, I focus on at-the-money, or ATM, straddles around earnings announcements. The ATM strike is the strike closest to the stock price before earnings. ATM straddles are commonly used to approximate the options market's expected move because they combine both upside and downside option prices.

The core research question is:

> Can pre-earnings option and liquidity features predict which earnings straddles are relatively overpriced or underpriced?

This project does not attempt to prove that earnings straddles are always profitable or always unprofitable. Prior research and market intuition already suggest that earnings options are often richly priced because investors demand protection or speculative exposure before earnings. Instead, this project focuses on the cross-section: among many stocks reporting earnings, can we rank which straddles are most overpriced and which are relatively less overpriced?

The key variable in the project is straddle edge:

straddle edge = realized move - implied move

The implied move is the move priced by the options market before earnings. The realized move is the stock's actual move after earnings. If edge is positive, the stock moved more than the straddle implied, meaning the straddle was relatively cheap. If edge is negative, the stock moved less than the straddle implied, meaning the straddle was expensive.

The final result is that earnings straddles are expensive on average, but the degree of overpricing is predictable. A simple out-of-sample linear or ridge regression model ranks straddles effectively. The predicted-best quintile loses much less than buying all straddles, while the predicted-worst quintile performs much worse. A theoretical long-best/short-worst spread remains positive even after simple bid/ask transaction-cost assumptions.

Methods

Data Construction

The project began by constructing a clean earnings straddle dataset. Each row represents one stock earnings event.

For each earnings event, the dataset links:

earnings announcement date
stock ticker
stock price before earnings
stock price after earnings
option prices before earnings
nearest option expiration after earnings
ATM call and put option prices

The main dataset uses the simple after-expiry straddle method. This means the option contract used is the first option expiration after the earnings announcement. Importantly, the option quote itself is observed before earnings. The word “after” refers to the expiration date being after the earnings event, not to the quote being observed after the event.

For example:

Earnings date: May 10
Option quote date: May 9
Option expiration date: May 17

The quote is known before earnings, but the option expires after earnings. This is useful because the option price should contain the market’s expectation of the earnings move.

The project originally also tested an isolated before/after expiry method, where the implied earnings move was estimated as:

after-expiry straddle - before-expiry straddle

However, this approach produced a much smaller sample and suspiciously low implied moves. The strict isolated dataset had only 329 usable rows, compared with over 7,000 strict rows for the simple after-expiry dataset. Because the isolated method was too restrictive and potentially biased, it was kept only as a robustness idea, while the simple after-expiry method became the main dataset.

The final simple dataset sizes were:

Base dataset: 9,804 events
Loose dataset: 9,022 events
Strict dataset: 7,412 events

The loose dataset keeps more observations and is used for main modeling. The strict dataset applies stronger liquidity filters and is used as a robustness check.

Implied Move

The main implied move variable is:

implied_move_simple = after_straddle_mid / stock price

The straddle midpoint price is:

after_straddle_mid = after_call_mid + after_put_mid

The call midpoint and put midpoint are calculated using the bid and ask prices:

call midpoint = (call bid + call ask) / 2
put midpoint = (put bid + put ask) / 2

For example, if a stock is trading at $100, the ATM call midpoint is $3, and the ATM put midpoint is $2.70, then:

straddle midpoint = $3 + $2.70 = $5.70
implied move = $5.70 / $100 = 5.70%

This means the options market is roughly pricing a 5.70% earnings move.

This feature was chosen because ATM straddle prices are a direct and interpretable way to estimate the market's expected move before earnings.

Realized Move

The realized move measures how much the stock actually moved after earnings. The main realized move uses the stock's next opening price relative to the previous close:

realized_move_open_to_prevclose = abs(next open price - previous close price) / previous close price

The open-to-previous-close move was chosen because it captures the immediate reaction to earnings. If earnings are released after market close, the next open reflects the overnight reaction. If earnings are released before market open, the same day's open reflects the immediate reaction.

A close-to-close version was also created as a secondary measure:

realized_move_close_to_prevclose

However, the main target uses the open reaction because it is closer to the event response and less affected by later intraday trading.

Straddle Edge

The main target variable is:

simple_straddle_edge_open = *realized_move_open_to_prevclose* - *implied_move_simple*

This measures whether the stock moved more or less than the options market implied.

If:

edge > 0

then the stock moved more than implied, so the straddle was cheap.

If:

edge < 0

then the stock moved less than implied, so the straddle was expensive.

For example:

implied move = 6%
realized move = 4%
edge = 4% - 6% = -2%

This means the straddle was expensive.

Another example:

implied move = 5%
realized move = 8%
edge = 8% - 5% = +3%

This means the straddle was cheap.

This target was chosen because it directly measures the economic question of the project: whether the option-implied move was too high or too low relative to the realized earnings move.

Liquidity Filters

Liquidity filters were applied because option prices can be noisy when contracts are illiquid. A high-quality straddle price should come from options with reasonable bid-ask spreads, open interest, and trading activity.

The key liquidity variables are:

after_call_spread_pct
after_put_spread_pct
after_call_oi
after_put_oi
after_call_vol
after_put_vol

The bid-ask spread percentage is:

spread percentage = (ask - bid) / midpoint

A small bid-ask spread means the option quote is more reliable. A large spread means the midpoint may not reflect a realistic tradable price.

For example, an option quoted at:

bid = $2.90
ask = $3.10
midpoint = $3.00
spread = $0.20
spread percentage = 6.67%

is much more reliable than an option quoted at:

bid = $1.00
ask = $5.00
midpoint = $3.00
spread = $4.00
spread percentage = 133.33%

Open interest measures how many contracts are currently open. Higher open interest usually suggests the contract is more active. Volume measures how many contracts traded on the quote date. Volume is stricter than open interest because an option may have open interest but no trades on a particular day.

Two filtered datasets were created:

Loose dataset:

- implied move > 0 and < 100%
- realized move >= 0 and < 100%
- call and put spread percentage < 75%
- call and put open interest >= 1

Strict dataset:

- implied move > 0 and < 100%
- realized move >= 0 and < 100%
- call and put spread percentage < 50%
- call and put open interest >= 10
- call and put volume >= 1

The loose dataset was used for main exploration and modeling because it preserves more observations. The strict dataset was used as a robustness check because it keeps only more liquid options.

The final row counts were:

Base: 9,804 events, 480 tickers
Loose: 9,022 events, 474 tickers
Strict: 7,412 events, 457 tickers

The loose and strict datasets gave similar average results, which suggests the main conclusion is not driven only by poor option quotes.

Features Created for Modeling

Several features were created to explain or predict straddle edge.

implied_move_simple

This is the market-implied move from the ATM straddle. It was the most important feature because Part 2 showed that high implied-move straddles tend to have worse future edge.

total_spread_pct
total_spread_pct = (after_call_spread_pct + after_put_spread_pct) / 2

This measures the average bid-ask spread of the call and put. It captures liquidity and quote reliability. It was chosen because wider spreads indicate noisier and more expensive-to-trade options.

total_open_interest
total_open_interest = after_call_oi + after_put_oi

This measures the total number of outstanding call and put contracts. It was included as a liquidity feature.

total_volume
total_volume = after_call_vol + after_put_vol

This measures the total number of call and put contracts traded on the quote day. It was included because higher trading activity may indicate more reliable market pricing.

log_total_open_interest
log_total_open_interest = log(1 + total_open_interest)

Open interest can be very uneven across stocks. Some contracts have only a few open contracts, while others have thousands. The log transformation reduces the effect of extreme values and makes the feature easier for a regression model to use.

log_total_volume
log_total_volume = log(1 + total_volume)

This has the same motivation as log open interest. Volume can vary widely, so the log transformation reduces extreme scale differences.

log_stock_price
log_stock_price = log(stock price)

Stock price was included because Part 2 showed a weak relationship between stock price and edge. Lower-priced stocks tended to have worse straddle edge. The log transformation was used because stock prices vary widely.

earnings_month

This is the calendar month of the earnings announcement. It was included to capture possible earnings-season effects. Earnings reports cluster in certain months, and pricing behavior may differ across earnings seasons.

Modeling Approach

Part 2: Baseline Quintile Tests

Before using regression models, the project tested simple one-variable signals. Each signal was sorted into quintiles:

Q1 = lowest values of the signal
Q5 = highest values of the signal

Then the average future straddle edge was calculated in each quintile.

The main variables tested were:

implied_move_simple
total_spread_pct
log_total_open_interest
log_total_volume
Stock price

The strongest single-variable signal was *implied_move_simple*.

In the loose dataset, sorting by implied move produced:

Q1 average edge = -0.56%
Q5 average edge = -2.54%
Q5 - Q1 = -1.98%

In the strict dataset:

Q1 average edge = -0.50%
Q5 average edge = -2.46%
Q5 - Q1 = -1.96%

This means high implied-move straddles are much more overpriced than low implied-move straddles.

The pattern was stable by year:

2018: Q5 - Q1 = -1.49%
2019: Q5 - Q1 = -1.39%
2020: Q5 - Q1 = -3.56%
2021: Q5 - Q1 = -1.82%
2022: Q5 - Q1 = -2.85%
2023: Q5 - Q1 = -1.11%
2024: Q5 - Q1 = -1.20%

This was important because it showed that the implied-move effect was not driven by only one year.

Other baseline findings were:

Higher total spread percentage → worse edge
Higher volume → better edge
Open interest alone → weak relationship
Higher stock price → slightly better edge

The correlations with future edge in the loose dataset were:

implied move correlation: -0.239
spread correlation: -0.073
open interest correlation: 0.001
volume correlation: 0.090
stock price correlation: 0.024

These results motivated the combined signal and regression models in Part 3.

Part 3: Combined Score and Regression Models

Part 3 moved from single-variable tests to combined prediction.

First, a hand-built overpriced score was created using the Part 2 findings:

overpriced_score =
rank_implied_move
+ rank_spread
- rank_volume
- rank_stock_price

Percentile ranks were used so that variables with different scales could be combined. Higher implied move and higher spread were treated as signs of overpricing. Higher volume and higher stock price were treated as signs of relatively better edge.

The hand-built score produced a clear monotonic pattern.

Loose dataset:

Q1 edge = -0.61%
Q2 edge = -0.94%
Q3 edge = -1.25%
Q4 edge = -1.83%
Q5 edge = -2.58%
Q5 - Q1 = -1.97%

Strict dataset:

Q1 edge = -0.64%
Q5 edge = -2.48%
Q5 - Q1 = -1.84%

This showed that a simple combination of implied move and liquidity features could rank overpriced straddles.

Next, out-of-sample regression models were trained. Two simple and explainable models were used:

linear regression
ridge regression

Complex machine learning models were not used because the goal was to keep the project interpretable and avoid overfitting. The dataset has around 9,000 observations, and the features are relatively simple. A linear model is easier to explain and directly shows which features predict better or worse edge.

The target was:

simple_straddle_edge_open

The regression features were:

implied_move_simple
total_spread_pct
log_total_open_interest

log_total_volume
log_stock_price
earnings_month

The models were tested using expanding-window out-of-sample validation:

Train 2018–2020 → Test 2021
Train 2018–2021 → Test 2022
Train 2018–2022 → Test 2023
Train 2018–2023 → Test 2024

This was chosen to avoid look-ahead bias. The model only used past data to predict future earnings events.

The linear regression performance was:

2021: RMSE = 3.62%, MAE = 2.27%, R^2 = 0.0588
2022: RMSE = 4.10%, MAE = 2.93%, R^2 = 0.1319
2023: RMSE = 4.10%, MAE = 2.81%, R^2 = -0.0153
2024: RMSE = 4.41%, MAE = 2.98%, R^2 = 0.0233

The ridge regression results were almost identical.

The R^2 values are small, which is normal in noisy financial prediction problems. More importantly, the predictions had a positive out-of-sample correlation with actual edge:

linear regression correlation = 0.248
ridge regression correlation = 0.248

This indicates that the models had useful ranking power.

When predicted edge was sorted into quintiles:

Q1 actual edge = -2.54%
Q2 actual edge = -1.62%
Q3 actual edge = -1.35%
Q4 actual edge = -0.79%
Q5 actual edge = -0.51%

Thus, the model successfully ranked straddles from worst to best. The predicted-best quintile was still slightly negative on average, but it was much less negative than the predicted-worst quintile.

The Q5 minus Q1 spread was:

linear regression: +2.03%
ridge regression: +2.03%
Baseline implied-move-only signal: +1.74%

This shows that the combined regression model slightly improved on implied move alone.

The ranking was positive in every test year:

2021: Q5 - Q1 = +2.38%
2022: Q5 - Q1 = +3.02%
2023: Q5 - Q1 = +1.29%
2024: Q5 - Q1 = +1.29%

The ridge coefficient summary showed:

implied_move_simple: -0.010130
log_total_open_interest: -0.003469
total_spread_pct: -0.001829
log_stock_price: -0.000014
earnings_month: 0.001077
log_total_volume: 0.004578

The signs are intuitive. Higher implied move predicts worse edge. Wider spreads predict worse edge. Higher volume predicts better edge.

Part 4: Strategy Evaluation and Transaction Costs

Part 4 evaluated the economic meaning of the model rankings. It did not train a new model.

The Part 3 predictions were merged back with option bid/ask data to create transaction-cost-aware measures.

The following measures were created:

mid_implied_move = after_straddle_mid / stock price
ask_implied_move = (call ask + put ask) / stock price
bid_implied_move = (call bid + put bid) / stock price

From these, three edge measures were created:

mid_edge = realized move - mid implied move
long_ask_edge = realized move - ask implied move

short_bid_edge = bid implied move - realized move

mid_edge is the research midpoint result.

long_ask_edge is more realistic for buying straddles because a buyer pays the ask.

short_bid_edge is a simplified short-straddle proxy because a seller receives the bid.

The average transaction-cost summary was:

average midpoint implied move = 5.79%
average ask implied move = 6.08%
average bid implied move = 5.49%
average realized move = 4.43%

average midpoint edge = -1.36%
average long-at-ask edge = -1.66%
average short-at-bid edge = +1.06%

The average straddle bid-ask spread was:

0.59% of stock price
10.59% of midpoint straddle value

This shows that bid-ask costs are meaningful and should not be ignored.

The main strategy results were:

Buy all straddles - midpoint: -1.36%
Buy all straddles - at ask: -1.66%

Buy Q5 predicted-best - midpoint: -0.48%
Buy Q5 predicted-best - at ask: -0.60%

Buy Q1 predicted-worst - midpoint: -2.43%
Short Q1 predicted-worst - at bid: +1.89%

Long Q5 / Short Q1 midpoint spread: +1.95%
Long Q5 at ask / Short Q1 at bid: +1.29%

These results show that buying all earnings straddles is unattractive on average. Buying only the predicted-best quintile loses much less, but it is still slightly negative after ask costs. The strongest result is the cross-sectional spread between predicted-best and predicted-worst straddles.

The transaction-cost-aware long Q5 / short Q1 spread remained positive:

+1.29%

Year-by-year results were also positive:

2021: midpoint spread = +2.38%, bid/ask spread = +1.61%
2022: midpoint spread = +3.02%, bid/ask spread = +2.16%
2023: midpoint spread = +1.29%, bid/ask spread = +0.79%
2024: midpoint spread = +1.29%, bid/ask spread = +0.76%

A top-decile test showed that the strongest signal is concentrated in the extremes:

D1 midpoint edge = -3.08%
D10 midpoint edge = -0.33%
D10 - D1 midpoint spread = +2.75%

D10 long ask edge = -0.43%
D1 short bid edge = +2.49%
Long D10 ask / Short D1 bid spread = +2.07%

This shows that the most extreme predicted groups have the strongest separation.

Results

The first major result is that earnings straddles are expensive on average.

In the loose dataset:

average implied move = 5.71%
average realized move = 4.27%
average edge = -1.44%

In the strict dataset:

average implied move = 5.73%
average realized move = 4.37%
average edge = -1.36%

The fact that the result is similar in the loose and strict datasets suggests that the average overpricing result is not solely driven by illiquid option quotes.

The second major result is that high implied-move straddles are especially overpriced. In the loose dataset, the lowest implied-move quintile had average edge of -0.56%, while the highest implied-move quintile had average edge of -2.54%. This pattern also appeared in the strict dataset and in every year from 2018 to 2024.

The third major result is that combining implied move and liquidity features improves ranking. The out-of-sample regression model generated a Q5 minus Q1 spread of approximately +2.03%, compared with +1.74% for the implied-move-only baseline.

The fourth major result is that the strategy interpretation remains meaningful after simple transaction cost assumptions. The long Q5 at ask / short Q1 at bid spread was +1.29% on average and positive in every test year from 2021 to 2024.

However, the predicted-best long-only straddles were still slightly negative after ask costs:

Buy Q5 predicted-best at ask = -0.60%

This means the model improves selection but does not fully overcome the general richness of earnings straddles on the long side.

Discussion

The results suggest that earnings straddles are generally expensive, but the degree of overpricing varies predictably across stocks. The strongest predictor is the implied move itself. When the options market prices an unusually large earnings move, the stock does tend to move more, but not enough to justify the high straddle price on average. This creates worse future edge for high implied-move straddles.

Liquidity also matters. Wider bid-ask spreads are associated with worse edge, likely because illiquid option prices are noisier and because trading frictions are higher. Higher option volume is associated with better edge, suggesting that more actively traded earnings options may provide more reliable and less overpriced exposure.

The regression model is intentionally simple. Linear and ridge regression were chosen because they are interpretable and reduce overfitting risk. The coefficient signs align with the earlier quintile tests, which increases confidence in the model's behavior. Complex machine learning models were not necessary for the main project because the simple model already produced stable out-of-sample ranking results.

The most important interpretation is that the model is not a naive long-straddle profit machine. Buying all straddles loses money on average, and even buying the predicted-best quintile at ask remains slightly negative. Therefore, the better interpretation is that the model is a cross-sectional ranking and filtering tool. It can help identify which earnings straddles are most overpriced and should be avoided by long-volatility traders. It can also support a theoretical relative-value volatility strategy: long the predicted-best straddles and short the predicted-worst straddles.

The short side must be treated carefully. Short straddles have tail risk, margin requirements, and potentially large losses if the stock moves sharply. The *short_bid_measure* measure used here is a simplified move-based proxy, not a full options PnL calculation. Therefore, the long-short results should be interpreted as evidence of relative mispricing, not as a fully implementable trading strategy without further risk controls.

There are several limitations. First, the project uses move-based edge rather than full option payoff PnL. A future extension should compute actual straddle returns based on option payoff at expiration or immediately after earnings. Second, the transaction-cost assumptions use bid and ask prices, but actual execution could vary depending on order size, market conditions, and liquidity. Third, the current feature set is intentionally simple. Additional features such as recent realized volatility, prior earnings moves, analyst forecast dispersion, market volatility, sector controls, market capitalization, implied volatility skew, and volatility term structure could improve the model.

Despite these limitations, the project produces a clear and defensible result. It shows that earnings straddles are expensive on average, but their relative overpricing is predictable using simple pre-earnings option and liquidity features. The result is stable across models, liquidity filters, transaction-cost assumptions, and test years.

Conclusion

This project developed a cross-sectional earnings straddle mispricing model using pre-earnings option prices and liquidity features. The main finding is that earnings straddles are expensive on average, with implied moves exceeding realized moves by roughly 1.4 percentage points in the main clean sample.

However, the overpricing is not uniform. High implied-move and wide-spread straddles are especially expensive, while higher-volume straddles tend to have better edge. A simple out-of-sample linear or ridge regression model ranks future straddle edge effectively. The predicted-best quintile outperforms the predicted-worst quintile by about 2.0 percentage points at midpoint and about 1.3 percentage points after simple bid/ask assumptions.

The final interpretation is that the model is most useful as a relative-value ranking and trade-filtering tool. It helps identify the most overpriced long-volatility trades to avoid and supports further research into long-best/short-worst volatility spreads. It should not be interpreted as proof that buying earnings straddles is profitable on average.